The Geometric Unitary Kudla Conjecture

Automatic convergence of Hermitian Fourier–Jacobi series

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Special Cycles

The Kudla Program

Relates special cycles on Shimura varieties to automorphic forms.

Kudla–Millson '90: At the level of cohomology this relation is explained by theta correspondence.

Kudla '97: The geometric Kudla Conjecture refines it to algebraic cycles.

Generating series with values in a Chow group are modular forms.

Hermitian lattice

Let E/F be a CM field; can take $E/\mathbb{Q}$ imaginary quadratic.

Let L be a Hermitian $\mathscr{O}_E$-lattice with signatures

$$(p,1) \text{ at one infinite place,} \qquad (p+1,0) \text{ at all others.}$$

We let

$$L_{\mathbb{Q}} := L \otimes_{\mathbb{Z}} \mathbb{Q}, \quad L_{\mathbb{R}} := L \otimes_{\mathbb{Z}} \mathbb{R}, \quad L_{\mathbb{R}}^{-} := L \otimes_{\mathscr{O}_E,\iota} \mathbb{C} = L \otimes_{\mathscr{O}_F,\iota} \mathbb{R}$$

for a indefinite place $\iota : E \hookrightarrow \mathbb{C}$ of L. We have a complex structure on $L_{\mathbb{R}}^{-}$ via this embedding.

Symmetric domain and arithmetic quotient

The symmetric domain is the space of negative lines

$$\mathrm{Gr}^{-}(L_{\mathbb{R}}^{-}) := \left\{ W \subset L_{\mathbb{R}}^{-} : \dim_{\mathbb{C}} W = 1, \langle W, W \rangle < 0 \right\}.$$

It is a homogeneous space for the unitary group $\mathrm{U}_L(F_{\mathbb{R}})$.

For a neat, stable congruence subgroup $\Gamma \subset \mathrm{U}_L(\mathcal{O}_F)$, the arithmetic quotient

$$\Gamma \backslash \mathrm{Gr}^{-}(L_{\mathbb{R}}^{-})$$

is a connected component of a unitary Shimura variety.

Example: Symmetric space

For the signature $(p, 1)$ diagonal form

$$\langle x, x\rangle = |x_1|^2 + \cdots + |x_p|^2 - |x_{p+1}|^2,$$

a negative line can be represented by $(z_1, \ldots, z_p, 1)$ with

$$|z_1|^2 + \cdots + |z_p|^2 < 1.$$

The domain of negative lines is a complex ball.

Connection to classical modular forms

If $p = 1$, the complex disc it is biholomorphic to the upper half-plane.

We recover elliptic modular forms.

If $p = 2$, we recover Picard modular forms.

Example: Divisors on the symmetric space

Take the diagonal form $|x_1|^2 + |x_2|^2 - |x_3|^2$ and lattice $L = \mathscr{O}_E^3$.

The orthogonal complement of $\lambda = (1,0,0)$ is

$$\lambda^{\perp} = \left\{(0, x_2, 1) \,:\, |x_2|^2 < 1\right\} \subset \mathrm{Gr}^{-}(L_{\mathbb{R}}^{-}),$$

a complex disc associated with $|x_2|^2 - |x_3|^2$.

Example: Rational Hermitian divisors

The single subball

$$\{(0, x_2, 1) : |x_2|^2 < 1\} \subseteq \{(x_1, x_2, 1) : |x_1|^2 + |x_2|^2 < 1\}$$

does not descend to the arithmetic quotient.

The union over the Γ-orbit of $\lambda^\perp$ does:

$$\bigcup_{\gamma \in \Gamma} \gamma(\lambda^\perp) = \bigcup_{\gamma \in \Gamma} (\gamma\lambda)^\perp \subset \Gamma \backslash \mathrm{Gr}^-(L_{\mathbb{R}}^-).$$

Rational Hermitian cycles

Given a tuple $\lambda = (\lambda_1, \ldots, \lambda_g) \in L_{\mathbb{Q}}^g$, $L_{\mathbb{Q}} = L \otimes_{\mathbb{Z}} \mathbb{Q}$, define

$$\mathrm{Z}(\lambda) := \left\{ W \in \mathrm{Gr}^-(L_{\mathbb{R}}^-) \,:\, W \perp \lambda_i \text{ for all } i \right\}.$$

This cycle is a smaller unitary symmetric domain or empty. If the span of the λ_i is not positive semi-definite, it is empty.

On $\Gamma\backslash\mathrm{Gr}^-(L_{\mathbb{R}}^-)$ we use the Γ-orbit of λ. For $[\lambda] \in \Gamma\backslash L_{\mathbb{Q}}^g$ define

$$\mathrm{Z}([\lambda]) := \bigcup_{\lambda' \in [\lambda]} \mathrm{Z}(\lambda').$$

Moment matrix and discriminant form

We group cycles by the image of λ in $\operatorname{disc}(L)^g$ for the discriminant form

$$\operatorname{disc}(L) := L^\vee / L, \quad L^\vee := \big\{ \lambda \in L_{\mathbb{Q}} \,:\, \operatorname{trace}_{E/\mathbb{Q}} \langle \lambda, \lambda' \rangle \in \mathbb{Z} \text{ for all } \lambda' \in L \big\}.$$

And by the moment matrix

$$t(\lambda) := \tfrac{1}{2} \big(\langle \lambda_i, \lambda_j \rangle \big)_{i,j} \in \mathrm{Mat}_g^\dagger(E),$$

where $\mathrm{Mat}_g^\dagger(E) := \big\{ t \in \mathrm{Mat}_g(E) \,:\, t = {}^\dagger t \big\}$, ${}^\dagger t = {}^{\mathrm{t}}\overline{t}$.

In codimension one this is just the square norm of a vector.

Special cycles

For a discriminant class $\mu \in \mathbf{disc}(L)^g$ and a moment matrix $t \in \mathrm{Mat}_g^{\dagger}(E)$, the special cycle is the formal sum

$$\mathrm{Z}(\mu; t) := \sum_{\substack{\lambda \in L_{\mathbb{Q}}^g \\ \lambda \equiv \mu, t(\lambda) = t}} \mathrm{Z}(\lambda).$$

It descends to the arithmetic quotient, since Γ is stable.

The presence of the discriminant class is related to the Weil representation.

Kudla's generating series

The Chow-valued generating series is

$$\theta_L^{\mathrm{Kudla}}(\tau) := \sum_{\mu \in \mathrm{disc}(L)^g} \mathfrak{e}_\mu \sum_t \big(\mathrm{Z}(\mu; t) \cap \omega^{g - \mathrm{rk}(t)} \big)\, e(t\tau),$$

where $e(t\tau) = \exp(2\pi i\, \mathrm{trace}_{F/\mathbb{Q}}(\mathrm{trace}(t\tau)))$, which depends on a choice of complex places of E.

We force codimension g by intersecting with the hyperplane; ω is the dual of the Hodge bundle.

A priori this is only a formal Fourier series.

Kudla's Conjecture

Conjecture: The Kudla generating series is a Hermitian Hilbert modular form of weight $p+1$ for the incoherent Weil representation.

$$\theta_L^{\mathrm{Kudla}} \in \mathrm{M}_{p+1}\big(\widetilde{\rho}_L^{(g)}\big) \otimes \mathrm{CH}^g\big(\Gamma\backslash \mathrm{Gr}^-(L_{\mathbb{R}}^-)\big).$$

Chow-valued modularity

The coefficients of $\theta_L^{\mathrm{Kudla}}$ lie in $\mathrm{CH}^g\big(\Gamma\backslash\mathrm{Gr}^-(L_{\mathbb{R}}^-)\big)_{\mathbb{C}}$.

This Chow group is not known to be finite dimensional.

So modularity is first asserted after applying linear functionals:

$$\ell\big(\theta_L^{\mathrm{Kudla}}\big) \quad \text{should be a Hermitian Hilbert modular form.}$$

Once modularity is proved, finite dimensionality for modular forms gives meaningfully

$$\theta_L^{\mathrm{Kudla}} \in \mathrm{M}_{p+1}\big(\widetilde{\rho}_L^{(g)}\big) \otimes \mathrm{CH}^g\big(\Gamma\backslash\mathrm{Gr}^-(L_{\mathbb{R}}^-)\big).$$

The divisor case

For $g = 1$ we obtain special divisors. Their generating series has the form

$$\theta_L^{\mathrm{Kudla}}(\tau) := \mathfrak{e}_0\,\omega + \sum_{\mu} \mathfrak{e}_\mu \sum_{t>0} \mathrm{Z}(\mu;t)\, e(t\tau),$$

where $e(t\tau) = \exp(2\pi i\, \mathrm{trace}_{F/\mathbb{Q}}(t\tau))$.

Theorem (Liu; Hofmann for $F = \mathbb{Q}$): This divisor generating series is modular.

Main Result

Geometric unitary Kudla Conjecture

Theorem (R.): Let L be an integral Hermitian lattice over a CM-field with signature $(p,1)$ at one infinite place and $(p+1,0)$ at all others.

For every $g \geq 1$, the Kudla generating series is modular:

$$\theta_L^{\mathrm{Kudla}} \in \mathrm{M}_{p+1}\big(\widetilde{\rho}_L^{(g)}\big) \otimes \mathrm{CH}^g\big(\Gamma\backslash \mathrm{Gr}^-(L_{\mathbb{R}}^-)\big).$$

Arithmetic inner products

Li–Liu construct arithmetic cycles and prove an arithmetic inner product formula relating them to derivatives of L-series akin to Gross–Zagier.

Their construction assumes modularity of the Chow-valued Kudla generating series: They take the Petersson inner product against it.

The theorem verifies this modularity hypothesis.

Formal Fourier–Jacobi series

The divisor case is known by Liu, following an argument of Yuan–Zhang–Zhang.

Higher codimension was already set up formally by Liu, following a strategy by Zhang.

The missing step is convergence:

symmetric formal Fourier–Jacobi series

are

Hermitian Hilbert modular forms.

Formal Fourier–Jacobi Series

Hermitian upper half space

Recall that E/F is a CM field; could take $E/\mathbb{Q}$ for simplicity, but will assume that $F \neq \mathbb{Q}$ to avoid the growth condition.

The Hermitian Hilbert upper half space is

$$\mathbb{H}_{E,g} := \left\{\tau \in \mathrm{Mat}_g(E \otimes_{\mathbb{Q}} \mathbb{R}) : \mathrm{Im}(\tau) > 0\right\}, \quad \mathrm{Im}(\tau) = \tfrac{1}{2i}(\tau - {}^{\dagger}\tau).$$

The special unitary group $\mathrm{SU}_{g,g}(\mathscr{O}_F)$ acts by

$$\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)\tau = (a\tau + b)(c\tau + d)^{-1}.$$

Weights and types

For weights $k \in \mathbb{Z}$, we have a slash action

$$\big(f|_k \gamma\big)(\tau) := \mathrm{norm}_{F/\mathbb{Q}}\big(\det(c\tau + d)^{-k}\big) f\big((a\tau + b)(c\tau + d)^{-1}\big).$$

Arithmetic types ρ are finite dimensional, complex representations of $\mathrm{SU}_{g,g}(\mathscr{O}_F)$ whose kernel is a congruence subgroup.

Hermitian modular forms

A Hermitian Hilbert modular form of weight $k \in \mathbb{Z}$ and type ρ is a holomorphic function

$$f : \mathbb{H}_{E,g} \longrightarrow V(\rho)$$

satisfying

$$(f|_k\gamma)(\tau) = \rho(\gamma) f(\tau) \quad \text{for all } \gamma \in \mathrm{SU}_{g,g}(\mathscr{O}_F).$$

It has a Fourier series expansion

$$f(\tau) = \sum_{t \in \mathrm{Mat}_g^\dagger(E)} c(f;t)\, e(t\tau).$$

Symmetry of Fourier coefficients

Fourier coefficients of a Hermitian Hilbert modular form satisfy

$$c(f;t[a]) = \mathrm{norm}_{F/\mathbb{Q}}(\det(\overline{a}))^k \, \rho\Big(\Big(\begin{smallmatrix} a & 0 \\ 0 & {}^{\mathrm{t}}\overline{a}^{-1} \end{smallmatrix}\Big)\Big)^{-1} c(f;t),$$

for $a \in \mathrm{GL}_g(\mathscr{O}_E)$ with $\det(a) = \det(\overline{a})$.

This is implied by invariance under the Siegel parabolic $P_g(\mathscr{O}_F) \subset \mathrm{SU}_{g,g}(\mathscr{O}_F)$.

Block decomposition

Fix $1 \le h < g$ and write

$$\tau = \begin{pmatrix} \tau_1 & {}^{t}z \\ w & \tau_2 \end{pmatrix} \in \mathbb{H}_{E,g},$$

where

$$\tau_1 \in \mathbb{H}_{E,g-h},\ \tau_2 \in \mathbb{H}_{E,h},\ z, w \in \mathrm{Mat}_{h,g-h}(E_{\mathbb{R}}).$$

Fourier indices decompose compatibly:

$$t = \begin{pmatrix} n & {}^{t}\overline{r} \\ r & m \end{pmatrix}.$$

Fourier–Jacobi expansion

A Hermitian Hilbert modular form has an expansion

$$f(\tau) = \sum_m \phi_m(\tau_1, z, w)\, e(m\tau_2).$$

The coefficients ϕ_m are Hermitian Hilbert–Jacobi forms.

Hermitian Hilbert–Jacobi forms

A Hermitian Hilbert–Jacobi form is a holomorphic function of

$$(\tau_1, z, w) \in \mathbb{H}_{E,g-h} \times \mathrm{Mat}_{h,g-h}(E_{\mathbb{R}}) \times \mathrm{Mat}_{h,g-h}(E_{\mathbb{R}}).$$

It transforms up to multiplication by $e(m\tau_2)$ under a Klingen parabolic in $\mathrm{SU}_{g,g}(\mathscr{O}_F)$.

We write their space as

$$\mathrm{J}^{(g-h)}_{k,m}(\rho).$$

Symmetric formal Fourier–Jacobi series

A symmetric formal Fourier–Jacobi series is a formal expression

$$f = \sum_m \phi_m(\tau_1, z, w)\, e(m\tau_2), \qquad \phi_m \in \mathrm{J}^{(g-h)}_{k,m}(\rho),$$

whose Fourier coefficients satisfy the same symmetry relation as a modular form.

Fourier coefficients are defined via the ones of Hermitian Hilbert–Jacobi forms.

We write this space as

$$\mathrm{FM}^{(g,h)}_k(\rho).$$

By Liu we have

$$\ell\left(\theta_L^{\mathrm{Kudla}}\right) \in \mathrm{FM}_{p+1}^{(g,g-1)}\left(\widetilde{\rho}_L^{(g)}\right) \quad \text{for all } \ell \in \mathrm{CH}^g\left(\Gamma\backslash\mathrm{Gr}^-(L_{\mathbb{R}}^-)\right)_{\mathbb{C}}^\vee.$$

We always have

$$\mathrm{M}_k^{(g)}(\rho) \subseteq \mathrm{FM}_k^{(g,h)}(\rho).$$

Is this an equality? Do symmetric formal Fourier–Jacobi series converge automatically?

Automatic Convergence

Main Result

Recall that we are in the Hermitian Hilbert modular forms setting with weights $k \in \mathbb{Z}$, arithmetic types ρ for the full modular group $\mathrm{SU}_{g,g}(\mathscr{O}_F)$.

Theorem (R.): For $1 \le h < g$, every symmetric formal Fourier–Jacobi series is convergent and modular.

Equivalently, for every weight k and arithmetic type ρ,

$$\mathrm{FM}_k^{(g,h)}(\rho) = \mathrm{M}_k^{(g)}(\rho).$$

Earlier automatic convergence theorems

Siegel modular forms genus 2, full level, no types: Aoki, Ibukiyama–Poor–Yuen.

Siegel modular forms genus 2 any type (Orthogonal Kudla Conjecture in codimension 2): Bruinier and R. independently using different methods.

Siegel modular forms arbitrary genus any type (Orthogonal Kudla Conjecture): Bruinier–R.

Hermitian modular forms any type, $\mathscr{O}_E/\mathbb{Z}$ norm Euclidean (Unitary Kudla Conjecture): Xia.

Concurrent work

Aoki–Ibukiyama–Poor treat genus 2 Siegel paramodular level.

Pollack proves cogenus $h = 1$ convergence in the Hermitian setting over imaginary quadratic fields.

Key steps of the proof

Trivial type: $\mathrm{FM}_\bullet^{(g,h)}$ is algebraic over $\mathrm{M}_\bullet^{(g)}$.

Trivial type, cogenus 1: Convergence on rational leaves of a fibration.

Convergence to meromorphic function on $\mathbb{H}_{E,g}$.

Control singularities via transversality to leaves of the fibration.

Reduction of arithmetic types (ample sheaf of modular forms) and cogenus h (formal theta decomposition).

Summary

Kudla's Conjecture and automatic convergence

Take the Shimura variety associated with a Hermitian lattice of signatures $(p,1),(p+1,0),\ldots,(p+1,0)$ over a CM-field.

Liu: Kudla's generating series is a symmetric formal Fourier–Jacobi series.

R.: Automatic convergence holds for these.

Li–Liu: Kudla's generating series yields arithmetic cycles that satisfy an inner product formula.

Details of the proof

Algebraicity

For fixed g and h one proves a dimension bound

$$\dim \mathrm{FM}_k^{(g,h)} \ll_{E,g} k^{[F:\mathbb{Q}]\, g^2}.$$

This matches the growth order of Hermitian modular forms:

$$\dim \mathrm{M}_k^{(g)} \asymp_E k^{[F:\mathbb{Q}]\, g^2}.$$

A transcendental formal series would force one extra power of k.

The dimension bound

Symmetry lets us reduce index m of ϕ_m by an action of $\mathrm{GL}_h(\mathscr{O}_E)$.

Minkowski theory controls the number of reduced m.

Symmetry successively dictates vanishing orders of ϕ_m that contribute to the dimension.

Hecke bounds for Hilbert–Jacobi forms bound m for which these ϕ_m may be nonzero.

Dimension estimates for remaining Hilbert–Jacobi forms via theta decomposition.

Torsion leaves

For cogenus 1 and $\alpha, \beta \in E^{g-1}$ define the leaf

$$\mathbb{H}_{E,g}[\alpha, \beta] = \left\{ z = \alpha\, {}^{\mathrm{t}}\tau_1 + \beta,\ w = \overline{\alpha}\, \tau_1 + \overline{\beta} \right\},$$

where as before

$$\tau = \begin{pmatrix} \tau_1 & {}^{\mathrm{t}}z \\ w & \tau_2 \end{pmatrix}.$$

Admissible rational choices of (α, β) are dense.

Restriction to a torsion leaf

On a torsion leaf, a Hermitian Hilbert–Jacobi form specializes to a Hermitian modular form of genus $g-1$.

$$\phi_m(\tau_1, z, w) \longmapsto \phi_m[\alpha, \beta](\tau_1) \in \mathrm{M}_k^{(g-1)}(N).$$

Reduction theory allows us to control its Fourier coefficients.

The growth estimate on leaves

For a cuspidal symmetric formal Fourier–Jacobi series

$$f = \sum_m \phi_m \, e(m\tau_2) \in \mathrm{FM}_k^{(g,1)},$$

we have

$$\|\phi_m[\alpha, \beta]\|_\infty \ll_f \mathrm{norm}_{F/\mathbb{Q}}(m)^{\frac{k-1}{2} + [F:\mathbb{Q}](g-1)}.$$

Convergence on leaves

On the leaf $\mathbb{H}_{E,g}[\alpha,\beta]$ one has the positivity condition

$$\mathrm{Im}(\tau_2) > \mathrm{Im}(\tau_1)[{}^{\mathrm{t}}\alpha].$$

The growth estimate is only polynomial in m, while $e(m\tau_2)$ gives exponential decay away from the boundary.

Therefore

$$\sum_m \phi_m(\tau_1, z, w)\, e(m\tau_2)$$

converges absolutely and locally uniformly on every torsion leaf.

From algebraicity to boundedness

Algebraicity gives a polynomial relation over $\mathrm{M}_\bullet^{(g)}$.

After multiplying by a leading coefficient, one obtains a monic relation

$$f^d + a_{d-1} f^{d-1} + \cdots + a_0 = 0.$$

On torsion leaves the series already converges.

The monic equation bounds the value of f by the modular coefficients a_i.

This gives local boundedness of partial sums.

Spreading convergence

The admissible torsion leaves are dense in $\mathbb{H}_{E,g}$.

On them the partial sums converge pointwise.

On compact subsets the same partial sums are locally bounded, after multiplying with a modular form as before.

Montel and Arzela–Ascoli imply locally uniform convergence on all of $\mathbb{H}_{E,g}$.

The limit is holomorphic, but we had to multiply by a modular form, so we obtain a meromorphic limit.

Modularity

The Fourier–Jacobi series gives invariance under a Klingen parabolic in $\mathrm{SU}_{g,g}(\mathscr{O}_F)$.

The symmetry condition gives invariance under the Siegel parabolic in $\mathrm{SU}_{g,g}(\mathscr{O}_F)$.

These subgroups generate $\mathrm{SU}_{g,g}(\mathscr{O}_F)$ by a K-theoretic computation.

Symmetric formal Fourier–Jacobi series are meromorphic Hermitian Hilbert modular forms.

Regularity

Assume there is an irreducible polar divisor.

Using Borel density and the complex structure, we see that it intersects one admissible leaf transversally.

On such a leaf the original formal series converges holomorphically.

By divisor pullback, the form is regular on this divisor.

Arithmetic types

The scalar theorem implies the vector-valued theorem.

Pair a vector-valued formal series with cusp forms for the dual representation.

By ampleness on the Satake compactification the latter span most fibres.

The pairings are scalar-valued symmetric formal Fourier–Jacobi series.

By the scalar theorem they are modular forms.

Higher cogenus

For $h > 1$ one reduces to cogenus $h - 1$.

Fixing a positive definite lower-right block m' gives a Fourier–Jacobi coefficient $\psi_{m'}$.

It admits a formal theta decomposition

$$\psi_{m'} = \sum_{\mu'} \widetilde{f}_{\mu'}\, \theta_{m',\mu'}.$$

The vector $(\widetilde{f}_{\mu'})_{\mu'}$ is a cogenus 1 symmetric formal Fourier–Jacobi series.